EXPERIMENT:3

Design a CPU scheduling program with C using First ComeFirst Served

Code:-

```
#include <stdio.h>
int main()
{
   int A[100][4];
   int i, j, n, total = 0, index, temp;
   float avg_wt, avg_tat;

   printf("Enter number of processes: ");
   scanf("%d", &n);

   printf("Enter Burst Time:\n");
   for (i = 0; i < n; i++)
   {
      printf("P%d: ", i + 1);
      scanf("%d", &A[i][1]);
      A[i][0] = i + 1;
   }
   for (i = 0; i < n; i++)
  {
        index = i;
 for (j = i + 1; j < n; j++)
      {
         if (A[j][1] < A[index][1])
            index = j;
```

```
        }
        temp = A[i][1];
        A[i][1] = A[index][1];
        A[index][1] = temp;
        temp = A[i][0];
        A[i][0] = A[index][0];
        A[index][0] = temp;
    }
    A[0][2] = 0;
   for (i = 1; i < n; i++)
    {
        A[i][2] = 0;
        for (j = 0; j < i; j++)
            A[i][2] += A[j][1];

        total += A[i][2];
    }
  avg_wt = (float)total / n;
    total = 0;
  printf("\nP\tBT\tWT\tTAT\n");
 for (i = 0; i < n; i++)
    {
        A[i][3] = A[i][1] + A[i][2];
        total += A[i][3];
  printf("P%d\t%d\t%d\t%d\n",
               A[i][0], A[i][1], A[i][2], A[i][3]);
    }
```

```
    avg_tat = (float)total / n;

    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

    return 0;
}
```

Output:

```
Enter number of processes: 3
Enter Burst Time:
P1: 8
P2: 5
P3: 3

P       BT      WT      TAT
P3      3       0       3
P2      5       3       8
P1      8       8       16

Average Waiting Time = 3.67
Average Turnaround Time = 9.00


...Program finished with exit code 0
Press ENTER to exit console.
```